

Sachin Shankar

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Objective

Passionate aerospace and rocketry enthusiast leveraging software development to push the boundaries of propulsion and flight systems. Seeking research, internship, or mentorship opportunities in aerospace engineering and mission-critical software, with long-term goals of contributing to next-generation launch vehicles at organizations like SpaceX or Boeing.

Education

Tesla STEM High School

9th Grade Student — STEM Focus

Sammamish, WA

2025 – Present

- Rigorous STEM curriculum with deep emphasis on science, technology, engineering, and mathematics.
- Active competitor in Science Olympiad, Science Bowl, and STEM academic competitions.

Awards & Achievements

★ 1st Place — Washington State Science Olympiad (WASO) 2026 2026

★ 8th Place — National Science Olympiad 2026 2026

★ 2nd Place — National Science Bowl (NSB) Regional Championship 2026 2026

★ 21-Time Medalist — Science Olympiad (Multiple Events & Seasons) 2023 – 2026

- Sustained multi-season excellence across a wide range of Science Olympiad events spanning physics, chemistry, engineering design, and earth science.

Projects

Solid Rocket Fuel Formulation & Engineering

Ongoing

Independent Research Project

- Independently researching and engineering novel solid rocket propellant compositions with a focus on performance, stability, and safety characteristics.
- Utilizing OpenRocket for flight simulations and trajectory modeling to validate propellant performance and predict vehicle behavior at target altitudes.

SuperWin — Windows Multi-Tool Utility

Ongoing

Solo Developer — C++

- Developing a native Windows desktop application in C++ featuring a suite of productivity tools including multi-level calculators with CAS (Computer Algebra System) engines, real-time desktop diagnostics, file format converters, and sleep/power management utilities.
- Architecting a modular multi-feature application using Win32 APIs with emphasis on performance, memory management, and low-level systems programming.

Space Odyssey — Discord Game Bot

Ongoing

Solo Developer — JavaScript

- Designed and built a fully-featured space-themed Discord game bot in JavaScript where players construct and manage space stations, compete for resources, and progress through game events.
- Integrated third-party image generation APIs to produce dynamic in-game visuals and built custom server workflows to automate game state management and event handling.

Technical Skills

Programming	Python (Upper-Intermediate), JavaScript (Upper-Intermediate), HTML/CSS (Upper-Intermediate), Java (Intermediate), C/C++ (Novice — Actively Learning)
Tools	OpenRocket, Git/GitHub, CAD Software (Novice — Actively Learning)
Mathematics	Calculus AB (Differential & Integral Calculus), Introductory Series & Sequences
Domains	Rocketry & Propulsion, Aerospace Systems, Software Engineering, Automation

Extracurriculars & Interests

- **Science Olympiad** — Multi-season competitor with 21 medals; experienced across events in physics, chemistry, engineering design, and earth science.
- **Rocketry** — Hands-on interest in solid propellant design, rocket motor engineering, and simulation-driven flight planning using OpenRocket.
- **Aerospace Engineering** — Passionate about applying software tools to accelerate propulsion research, avionics development, and launch vehicle design.
- **Competitive STEM** — Consistent top performer in science bowls, olympiads, and academic challenge circuits at regional and national levels.

Career Goals

Aspiring to pursue aerospace engineering at the university level and contribute to the commercial spaceflight industry. Particularly interested in propulsion systems, avionics software, and launch vehicle design — with SpaceX and Boeing as long-term target employers.